



**CENTRALE LYON
ENISE**



Finish Turning

Subtitle

Group n°3

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2. Introduction

The Tool Material Pair (TMP), as defined in the French standards NF E 66-520 is a technique or methodology used to define specific working conditions for a combination of the selected cutting tool, the workpiece material, the cutting condition along with the specified cutting fluid. The main purpose of this method is to try and unify the manufacturing procedure around the globe to help different manufacturers understand each other and operate with similar conditions to improve cross-platform integration in facilities.

This method studies all these parameters, focusing on the tools and material's overall performance, their compatibility to acquire an acceptable working domain for the combination of both. Our cutting operation of interest is *Finish* Turning with the details regarding the material and tool are specified in later parts of the report.

3. Experiment Overview

3.1. Objectives

- Verify that the provided machining setup is efficient and reliable; that is, the provided insert can perform the required finish turning operation on the material provided using the available tool holder and machine
- Identify the working domain of the provided machining setup
- Build fitting models based on measured data to predict process outputs when changing the inputs

3.2. Methodology

The **Tool–Material Pair (T.M.P.)** method provides a systematic five-step framework to verify that a specific cutting tool can perform a cutting job on a specific workpiece material safely, efficiently, and predictably. Rather than relying on assumptions, the T.M.P. method establishes proven operating limits, performance models, and wear behavior before full-scale production begins.

Step 1: Qualification

The qualification stage establishes a reliable baseline for the cutting process. Initially, the tool manufacturer's catalog is consulted to obtain recommended starting values for cutting speed (V_c), feed rate (f), and depth of cut (a_p). Although these values are not optimized, they provide a safe and reasonable starting point. Short-duration test cuts (approximately 10 seconds each) are then performed to verify three critical conditions:

-Chip morphology: Safe machining is indicated by short, broken chips (commonly "6-shaped" chips). The appearance of long, continuous chips or chip nests is unacceptable, as they pose safety risks and can interfere with machine operation.

-Process efficiency and surface roughness measurements: Experimental results, especially specific cutting pressure and surface roughness, are compared with theoretical predictions. If measured values align with theoretical expectations within $\pm 20\%$ deviation, the experimental setup is considered efficient.

-Process reliability: Experimental results should stay consistent in different tests. If measured values are within $\pm 5\%$ deviation of their average value, the experimental setup is considered reliable.

-Process stability: Cutting forces are measured during the test runs. Stable machining is confirmed when force signals remain consistent without excessive fluctuations. Significant force variation indicates vibration, poor rigidity, or improper setup, and testing must be halted.

Step 2: Determination of the Minimum Cutting Speed

Cutting at excessively low speeds is detrimental. At low cutting speeds, material does not shear cleanly and tends to adhere to the cutting edge, forming a built-up edge. To identify this lower limit, cutting speed is gradually reduced by 10% from its recommended value while monitoring the specific cutting pressure. A sudden increase in this pressure indicates the onset of unstable cutting conditions. The speed at which this spike occurs is defined as the minimum allowable cutting speed. Below this value, machining is prohibited. This limit is formally documented and enforced during process planning.

Step 3: Definition of the Working Domain (Segmentation Diagram)

The next step is to identify a safe and effective combination of feed rate and depth of cut. This is achieved using a segmentation diagram, which maps the acceptable operating region.

Two competing constraints are considered:

- Chip Formation Constraint

At very low feed rates, the chip breaker geometry on the insert becomes ineffective, resulting in long, hazardous chips.

- Surface Finish Constraint

At high feed rates, surface roughness increases beyond acceptable limits.

These constraints are plotted on a feed–depth graph. The overlapping region where both acceptable chip control and acceptable surface finish are achieved is defined as the working domain. Any parameter combination within this domain ensures stable and safe machining.

Step 4: Process Modeling and Prediction

Conducting physical experiments for every parameter change is inefficient. Therefore, a predictive mathematical model is developed. Experimental data from previous tests are compiled, and a mathematical relationship is established to calculate specific cutting pressure and surface roughness as functions of cutting speed (V_c), feed rate (f), and depth of cut (a_p). Model coefficients are optimized using numerical tools until the model accurately reproduces experimental results. Once calibrated, the model can reliably predict cutting forces, power consumption, and machine load for any parameter combination within the working domain.

Step 5: Tool Wear and Lifetime Modeling

Tool wear is inevitable, but only progressive flank wear is considered acceptable and predictable. Sudden chipping or catastrophic failure indicates a flawed process and is excluded from normal wear modeling. Wear tests are conducted to establish a Taylor tool life model, which relates tool life to cutting speed. This model allows estimation of usable tool life under different operating conditions.

Importantly, tool wear affects both power consumption and surface finish. A worn tool requires more cutting power and produces a rougher surface than a new tool. To address this, production planning models are conservatively calibrated by adding safety margins (for example, increasing predicted surface roughness by a defined percentage). This ensures that even the final part produced at the end of tool life remains within quality specifications.

4. Theoretical Background

4.1. Specific Cutting Pressure (K_c)

The value of K_c defines the specific amount of power required to remove the material at the specific flowrate. It acquires the units of force per square area [N/mm^2] and is mathematically expressed as:

$$K_c = \frac{\text{Cutting Power}}{\text{Work Material Flowrate}} = \frac{P_c}{V_c \cdot f \cdot a_p} \quad (1)$$

where P_c is the cutting power in watt, V_c is the cutting speed in m/min, f is the feed rate in mm/rev, and a_p is the depth of cut in mm. Together, a_p and f characterize the theoretical cross-sectional area of the chip (undeformed chip area). Multiplying this area with the cutting speed V_c provides the volumetric flowrate of material removed per minute.

There are three different forces acting on the material at the same time: cutting force F_c , feed force F_f and penetration force F_p . So, the cutting power is the combination of all these forces and their respective velocities.

$$P_c = V_c F_c + F_f V_f + V_p F_p \quad (2)$$

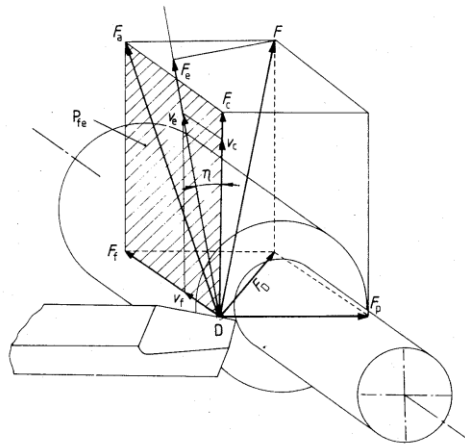


Figure 1: Directions of forces and velocities in a turning operation

As shown in the figure above, F_c and V_c are tangent to the tool, F_f and V_f are parallel along the axis of the workpiece and F_p and V_p are perpendicular to the job axis. As we know that in turning, usually $V_p = 0$ and $V_c \gg V_f$. So, equation (2) can be approximated as:

$$P_c = V_c F_c \quad (3)$$

So, replacing this value in equation (1) provides:

$$k_c = \frac{F_c(V_c)}{V_c \cdot f \cdot a_p} = \frac{F_c}{f \cdot a_p} \quad (4)$$

4.2. Surface Roughness (R_a)

The theoretical surface roughness (R_a) in μm can be calculated theoretically by the following formula:

$$R_a = \frac{f^2}{18\sqrt{3} \times r_a} \quad (5)$$

where r_a is the nose radius of the tool in mm, which is 0.4 mm.

5. Apparatus Involved

5.1. Machine and Lubrication Condition

The machine for our operation was 4-axis turning machine under dry conditions (no lubrication). The machine is manufactured by **CMZ**, a major equipment builder within Europe. Model number is **TC 25 Y**.

5.2. Profilometer

For the surface roughness measurements, we utilized a portable roughness tester **Mahr MarSurf PS10**. This device utilizes a tactile stylus method (the stylus runs over the microscopic peaks and valleys of the part) with a diamond stylus tip ($r_{tip}=2\mu m$). The standard transverse length (L_t) was 5.6mm. The cut-off wavelength (λ_c) was 0.8mm and the evaluation length comprised of 5 sample lengths ($5 \times \lambda_c = 4.0mm$). The raw data was filtered using a phase correct Gaussian filter in accordance with **ISO 16610-21** standard.

5.3. Sensor

The forces of interest (cutting force F_g , feed force F_f and penetration force F_p) were acquired using **Kistler Type 9129AA** three-component piezoelectric dynamometer. The platform has a measuring range of $\pm 10kN$, with a multichannel charge amplifier digitized for analysis with system linearity of $\leq \pm 0.3\%$.

5.4. Microscope for Tool Wear

Tool wear progression in the end was monitored using **Leica MZ12.5** high-resolution stereomicroscope. The microscope was periodically utilized during the last step of TMP to measure Flank Wear ($V_{B,max}$) on the inserts. The wear criteria were in accordance with the **ISO 3685** standards.

5.5. Workpiece (Bulk Material)

The work material utilized was Chromium-Molybdenum Alloy **42CrMo4** (AISI 4142). It lies in the medium carbon content category ($<0.8\%wt C$). Both the alloying elements significantly boost the steel's hardness, yield strength, wear rate and corrosion resistance. The Brinell hardness of 187-229 HB proposes that the material is either in a normalized or an annealed state. While this prevents the extreme wear in hard turning, the cutting temperatures generated are quite high because of high yield strength (900-1000 MPa), particularly in our case of dry machining. Also, the low thermal conductivity of this alloy tends to concentrate heat at the tool-material interface, resulting in enhanced crater wear (K_7) as material doesn't have enough time to conduct heat away from the zone of operation, also provoking build-up edge (BUE) at even lower cutting speeds, specified by the machinability rate of 65%. The chemical composition and mechanical properties of the work material are listed in the tables.

Elements	Content (wt%)
Carbon (C)	0.40-0.45
Silicon (Si)	0.15-0.35
Manganese (Mn)	0.75-1.00
Phosphorous (P)	≤ 0.035
Sulfur (S)	≤ 0.04
Chromium (Cr)	0.80-1.10
Molybdenum (Mo)	0.15-0.25

Table 1: Chemical composition of 42CrMo4

Property	Value
Hardness, Brinell	187-229
Hardness, Knoop	230
Hardness, Rockwell B	94
Hardness, Rockwell C	15
Hardness, Vickers	218
Modulus of Elasticity	206 GPa
Bulk Modulus	163 GPa
Poisson's Ratio	0.29
Machinability	65%
Shear Modulus	80 GPa

Table 2: Mechanical Properties of 42CrMo₄

5.6. Equipment Operating Procedure

1. Feed the program in the machine. For qualification, the fed parameters were the selected reference parameters after discussion with the professor.
2. Start the recording of the sensor to obtain forces readings, then start the machining operation
3. After the completion of the process, remove the workpiece for surface roughness inspection and remove the insert for inspection under microscope for flank wear evaluation.
4. Record all gathered data and put the workpiece and insert back.
5. Repeat the process 5 times under same conditions for qualification tests.
6. Repeat the similar steps for other stages of TMP with the only difference being the parameters set for the specific TMP stage by the supervisor.

6. Tool Description

6.1. Insert

The cutting tool specified for this operation is from the family **Coroturn 107** manufactured by **SANDVIK Coromant**. The name of the inset is **CCMT 09 T3 04-PF 4415**. This tool is specifically designed for finishing operations.



Figure 2: SANDVIK Coromant Coroturn 107 CCMT 09 T3 04-PF 4415

The full properties for the tool can be acquired both from the [SANDVIK's website](#), main geometric specifications are detailed below.

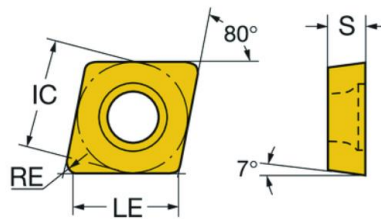


Figure 3: Insert geometry

Inscribed circle diameter (IC)	9.525 mm
Cutting edge effective length (LE)	0.9369 mm
Corner radius (RE)	0.3969 mm
Insert thickness(S)	3.9688 mm

Table 3:3 Main geometric measurements of cutting insert

The name of the tool provides all the essential details as follows:

- **C:** specifies the *shape* of the insert, in our case, it's a *rhomboid shape* with an angle of 80°
- **C:** describes the *relief angle* of tool, in our case which is 7°
- **M:** describes the *allowable tolerances* for the tool, which is ± 0.13 (on the thickness side)
- **T:** defines the *type of clamping and shape of chip breaker*, in our case, this is a *single sided countersunk (through hole) with screw clamping*.
- **09:** defines the *dimension for the cutting edge* which is $\sim 9.7\text{mm}$.
- **T3:** specifies the *thickness of the insert* which is 3.97mm .
- **04:** the *nose radius (tip radius)* which is about 0.4mm .
- **PF:** explains that this tool is designed specifically for *finishing operations* i.e., P=Steel and F=Finishing.
- **4415:** the detail about *material of insert*. In our case, it's a *hard carbide (HC)* base with a *multi-layered CVD* coating, comprising of. This implies that the tool is suitable for *medium-to-high-speed* steel machining operations with enhanced wear and heat resistance. (check appendix 1)

6.2. Tool Holder

The Tool holder used in the operations is **SCLCL 25 25 M 09**.



Figure 4: SCLCL 25 25 M 09 tool holder

Like the tool insert's case, we can capture all useful key characteristics of the holder from its code, in accordance with **ISO 1832**:

- **S:** defines the type of holder-insert connection, which is, in this case, a screw to join insert and holder
- **C:** specifies the shape of the insert, in our case, it's a rhomboid shape with an angle of 80°
- **L:** the angle of attack or approach angle (K_r) for the insert, which is $\sim 95^\circ$
- **C:** describes the relief angle of tool, which is 7° in our case
- **L:** defines the cutting direction of tool, which is left to right in our case (Left hand tool)
- **25 25:** defines the shank size of the holder i.e., 25 mm in both width and height respectively
- **M:** specifies the total length of the tool i.e., 150 mm
- **09:** it specifies that the holder can fit an insert with 9 mm edge length (*check appendix 2*)

7. Qualification

7.1. Process Stability

As shown in Figure 5, the sensor readings produce three plots displayed simultaneously. They are F_p (red), F_c (blue) and F_f (pink) respectively. For the process to be considered stable, small fluctuations in the force value is only acceptable for the duration of the process. The force is taken as the average between the start and end calculated by the software.

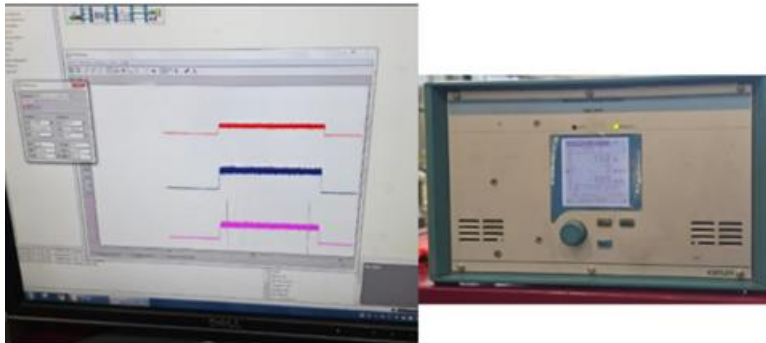


Figure 5: PC Interface (left) and Kistler recorder end (right)

7.2. Process Efficiency (K_c Evaluation)

It is the first stage of TMP, which is performed to determine whether the cutting tool is able to perform the cutting operation under reference parameters. This step lays down the foundation for the next steps of TMP analysis. It ensures that the cutting process would start within a stable operating zone, avoiding any failure at the start, and guides the next steps. The theoretical specific cutting pressure $K_c (theo)$ is used as the reference to determine whether the values of experimental specific cutting pressure $K_c (exp)$ in the 5 iterations are within the $\pm 20\%$ allowable range. In order to calculate $K_c (theo)$, we utilize the following formula:

$$h = f \cdot \sin(K_r) \quad (6)$$

where K_r is already specified earlier is the orientation angle of the insert, 95° , and for h , we compare it with the provided database of **SANDVIK Coromant** reference. The material in the handbook closest to our specification was **XC-80**.

Figure 6: Mémotech series handout for K_c (theo) turning

The reference cutting parameters are usually provided by the tool manufacturer. The initial parameters can be determined from both the back of the insert's casing and also from [SANDVIK's website](#). Although the manufacturer provided reference values for cutting parameters, after discussion with the professor and based on past experiments, the professor suggested different values as the new reference.

Quantity	Units	Reference Value (manufacturer)	Suggested Value (by Professor)
V_c	m/min	430 [465-375]	300
f	mm/rev	0.11 [0.06-0.22]	0.13
h	mm	0.11 [0.06-0.22]	Same as database
a_p	mm	0.3 [0.1-2]	Same as database

Table 4: Reference and suggested cutting parameters

Putting values in equation (6) yields,

$$h = 0.13 \times \sin(95^\circ) \approx 0.1295$$

Performing interpolation for obtaining the value of K_c (theo),

h (mm)	K_c (theo) (N/mm ²)
0.1	4800
0.2	3600
0.1295	4446

Table 5: Evaluation of K_c (theo) by interpolation

Cutting Speed (m/min)	Cutting Force (N)	K_c (exp) (N/mm ²)
300	172.17	2648.77

300	179.46	2760.92
300	178.16	2740.92
300	178.96	2753.23
300	178.41	2744.77

Table 6: Cutting force and K_c values for qualification tests

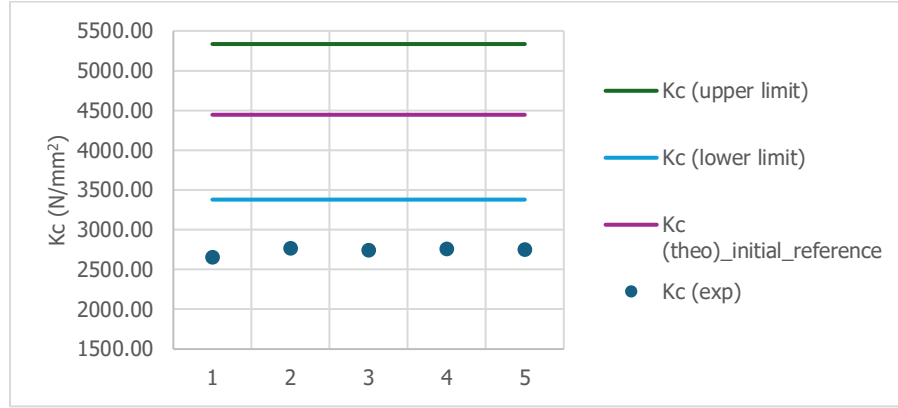


Figure 7: Plot of experimental K_c values against initial reference bounds for qualification tests

The experimental data after 5 tests were much lower than the lower bound i.e., 3556.8 N/mm², rendering the qualification step to fail. However, on discussion with the professor, who explained to us that this problem arises because sometimes the cutting tool, work material and testing conditions utilized by the manufacturer to develop the database are different from the ones performed in lab i.e., the cutting tool could be different based on the process selected for different production rates, the work material could be different due to any specific heat treatment process utilized during production and similar scenario with operational conditions.

Furthermore, in detailed discussion with our colleagues (Pedro CAVICCHIOLI, Sofia CILIA and Pedro MIRAGLIA, 2025), there was an important development in this specific behavior for K_c (theo). There was an observation that the work material used in lab is different from the reference steels in both the books of Memotech. This explanation can justify that such a high value for K_c (theo). Thus, a different reference was utilized to calculate the value for K_c (theo) from the SANDVIK Coromant database. This new reference can be provided as,

$$K_c(\text{theo}) = K_{c1} \cdot h^{-m_c} \cdot \left(1 - \frac{\gamma_0}{100}\right) \quad (7)$$

where K_{c1} defines the normalized specific cutting force in N/mm², an empirical constant measuring the material's resistance at reference chip thickness of $h=1.0$ mm and neutral rake angle ($\gamma_0=0^\circ$); m_c is the chip thinning exponent (dimensionless quantity) modelling non-linear increase in K_c with decrease in chip thickness; and γ_0 is the effective rake angle, which is defined by both the rake angle of the insert (γ_i) and the holder (γ_h) to apply the correction factor based on the geometry of the tool.

$$\gamma_0 = \gamma_i + \gamma_h \quad (8)$$

The values for both k_{c1} and m_c were taken from SANDVIK's handbook.

MC code	Material group	Material sub-group	Manufacturing process	Heat treatment	nom	Specific cutting force, K_{c1} (N/mm ²)	m_1
P1.1.2.AN	1	1	Z	AN annealed	125 HB	1500	0.25
P1.1.2.HT	1	1	Z	HT hardened+tempered	190 HB	1770	0.25
P1.2.2.AN	1	2	Z	AN annealed	190 HB	1700	0.25
P1.2.2.HT	1	2	Z	HT hardened+tempered	210 HB	1820	0.25
P1.3.2.AN	1	3	Z	AN annealed	190 HB	1750	0.25
P1.3.2.HT	1	3	Z	HT hardened+tempered	300 HB	2000	0.25
P1.4.2.AN	1	4	Z	AN annealed	220 HB	1180	0.21
P1.5.C.HT	1	5	C	HT untreated	150 HB	1400	0.25
P1.5.C.AN	1	5	C	AN hardened+tempered	300 HB	2880	0.25
P2.1.2.AN	2	1	Z	AN	175 HB	1700	0.25
P2.2.2.AN	2	2	Z	AN	240 HB	1950	0.25
P2.3.2.AN	2	3	Z	AN	260 HB	2020	0.25
P2.4.7.AN	2	4	Z	AN	275 HB		
P2.5.2.HT	2	5	Z	HT hardened+tempered	330 HB	2000	0.25
P2.6.C.UT	2	6	C	UT untreated	200 HB	1600	0.25
P2.6.C.HT	2	6	C	HT hardened+tempered	380 HB	3200	0.25
P3.0.2.AN	3	0	Z	AN annealed	200 HB	1950	0.25
P3.0.2.HT	3	0	Z	HT hardened+tempered	380 HB	3100	0.25
P3.0.C.UT	3	0	C	UT untreated	200 HB	1950	0.25
P3.0.C.HT	3	0	C	HT hardened+tempered	340 HB	3040	0.25
P3.1.2.AN	3	1	Z	AN annealed	250 HB	2360	0.25
P3.2.C.AQ	3	2	C	AQ annealed/quenched or annealed	300 HB	3000	0.25
P4.0.5.NS	4	0	S	NS not specified	150 HB		

Figure 8: Values of K_{c1} and m_c from SANDVIK database

However, there was no value proposed for our insert's rake angle γ_1 . Upon discussion, we agreed on utilizing the rake angle for the tool insert **CCMT 09 T3 04-PF 5015** (different material properties as 4415 but similar geometrical properties-our concern). So, $\gamma_1 = 7^\circ$ and for our tool holder, after checking the database again, the rake angle was neutral i.e., $\gamma_h = 0^\circ$. With these parameters, we have equation (8) as,

$$K_c (theo) = 1700 \times 0.1295^{-0.25} \times \left(1 - \frac{7}{100}\right) = 2636 \text{ N/mm}^2$$

This new value for $K_c (theo)$ is much closer to our experimentally obtained values after 5 qualification tests, with its allowable region encompassing all the data points of qualification.

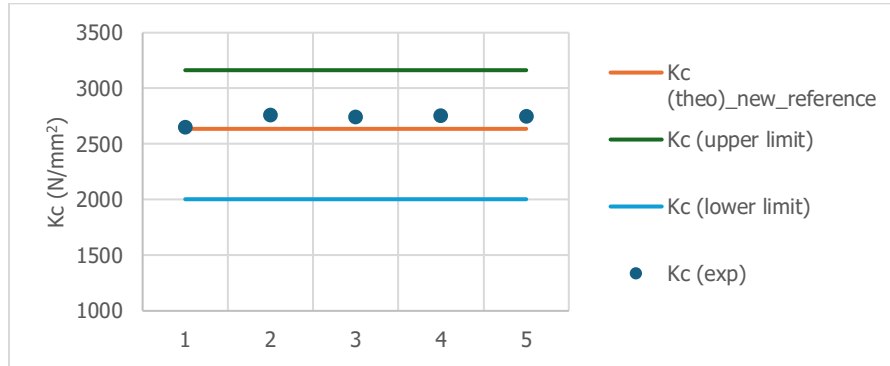


Figure 9: Plot of experimental K_c values against new reference bounds for qualification tests

It can be concluded that all specific pressure values for the qualification tests lie within the acceptable range around the theoretical value. Also, the force measurement plots on the sensor software were stable, indicating acceptable process stability.

7.3. Surface Roughness (R_a) Evaluation

Using equation (5), we can acquire the *theoretical surface roughness* $R_a (theo)$ as.

$$R_a (theo) = \frac{0.13^2}{18\sqrt{3} \times 4} \approx 1.355 \mu\text{m}$$

After each test, the workpiece was removed and examined by the profilometer to get the values of surface roughness. For each test, three values of surface roughness were obtained by rotating the workpiece by 120° and then calculating the mean of those readings. The results are tabulated in the table below.

Test	$R_{a,1}$	$R_{a,2}$	$R_{a,3}$	$R_{a,mean}$
1	0.997	1.06	0.973	1.01
2	1.001	0.932	1.017	0.98
3	0.945	0.964	0.933	0.95
4	0.902	0.958	0.941	0.93
5	0.913	0.903	0.917	0.91

Table 7: R_a values measured in the qualification tests

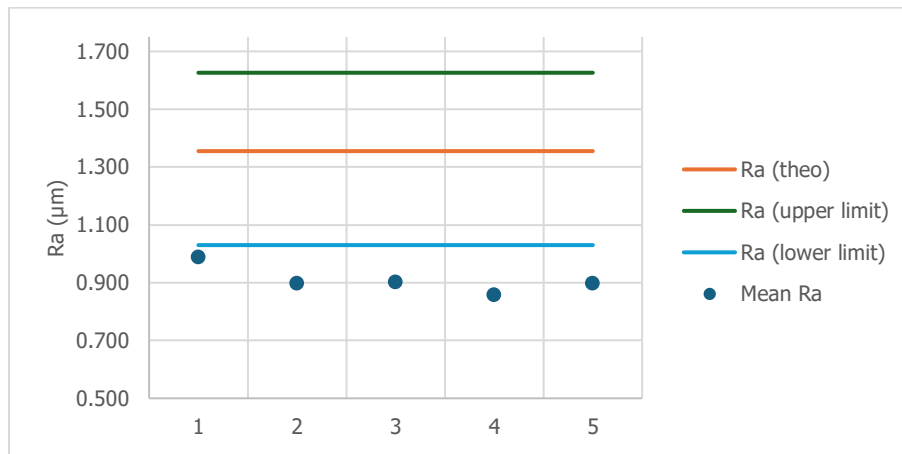


Figure 10: Plot of experimental R_a values against theoretical bounds for qualification tests

It was observed clearly that the values of surface roughness obtained from the qualification test were slightly lower than the lower bound of the theoretical surface roughness. This behavior of experimental roughness can be attributed to process instabilities like vibrations, tool runout, or material side flow (He et al., 2018). However, apart from these general probable reasons, there was another smoothing mechanism under play that was causing the experimental roughness to be lower than the allowable limit. This secondary reason can be related to the specific geometry of the tool insert. With a relief angle of 7° , the secondary cutting edge gives rise to a wiper effect. This can be further explained as in theoretical models, only the tip of the nose radius touches the material surface under operation. However, because of the material's elasticity and clearance angle of the tool, the trailing edge of the tool instead of cutting, rubs and plastically deforms the material's rough surface termed as burnishing. This smoothing effect is the major cause of our roughness values being lower than the theoretical model.

Another important observation was the continuous decrease of surface roughness after successive runs of qualification. This behavior can be attributed to the run-in period of the tool, in which the microscopic surface irregularities on the tool surface stabilize after usage (due to local flattening of cutting edge), resulting in a constantly smooth cutting action.

7.4. Chip Evaluation

The chip pictures were acquired from Group 1 on this turning operation for each iteration of the qualification test and they were evaluated against the chips evaluation table taught to us in the theoretical class.

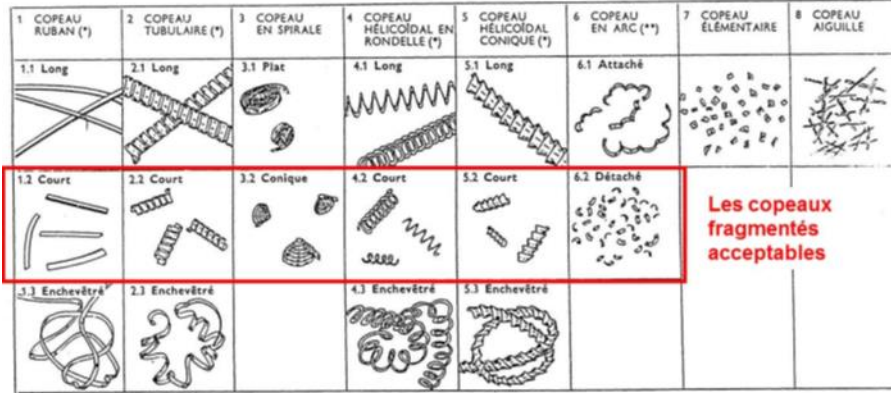


Figure 11: Chips geometry evaluation reference

The allowable standard for chips is that they should be short, segmented and shiny. Marked in Figure 11, we can see the type of chips that are allowable for the cutting operation, indicating its occurrence within the operating window.

We can categorize the obtained chips as.

Test 1	Test 2	Test 3	Test 4	Test 5
Category: 4.2	Category: 4.2	Category: 4.2	Category: 4.2	Category: 6.2

Table 8: Chips categorization for qualification tests

It can be observed that the color of the chips in all the tests was likely dark blue, indicating the generation of high temperatures at tool-material interface. This confirms that most of the thermal energy generated due to plastic deformation and friction was effectively transferred to chips, rather than its accumulation inside insert or material (resulting in unwanted thermal stresses). This high temperature provokes rapid and considerable oxidation of the chips on contact with atmospheric oxygen. While upper side developed a dark blue hue, the underside of the chip retains high brightness and smooth finish.

The overall qualification stage can be summarized in the form of a table shown below.

7.5. Process Reliability

The cutting operation is considered reliable if the experimental measurements from the five qualification tests fall within $\pm 5\%$ around their average value. This criterion serves not only as a qualification condition but also as a safety and consistency measure, confirming the suitability of the cutting tool for the selected work material under the given cutting conditions. It was tested from both K_c and R_a measurements as follows:

$$K_c(ave) = 2730 \text{ N/mm}^2$$

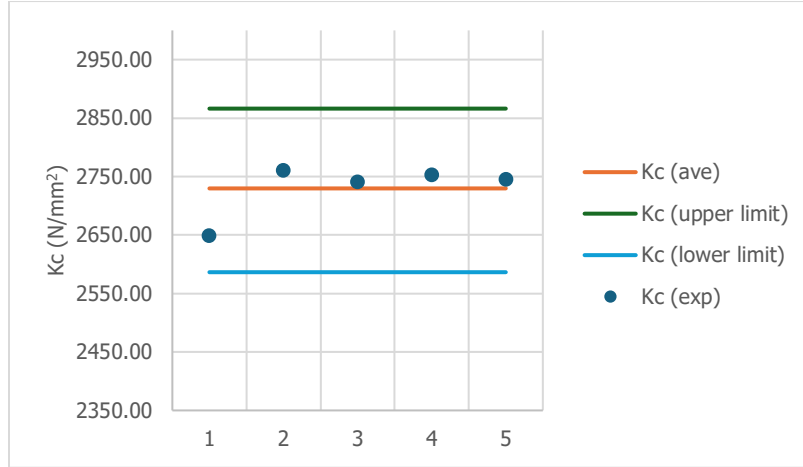


Figure 12: Distribution of experimental K_c values around their average ($\pm 5\%$ around the average value)

In the present study, the experimentally obtained K_c values for the provided cutting tool all lie within $\pm 5\%$ range around their average. This compliance confirms the mechanical stability and reliability of the cutting process.

$$R_a(ave) = 0.957 \mu m$$

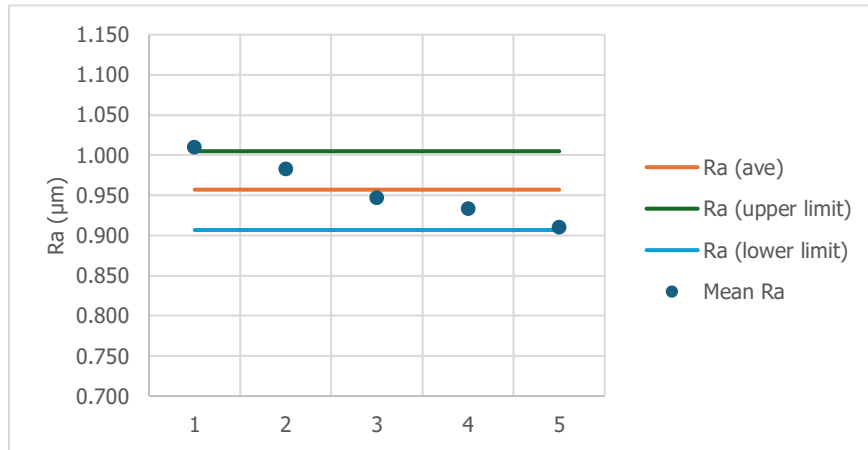


Figure 13: Distribution of experimental R_a values around their average ($\pm 5\%$ around the average value)

The measured R_a values for most machining passes consistently fall within the $\pm 5\%$ tolerance range around their average value. Only the first one is slightly above the upper limit at $\sim 5.5\%$ difference from the average value. This confirms the reliability of the process from surface roughness point of view. The results indicate stable chip formation, controlled tool–workpiece interaction, and the absence of adverse effects such as excessive built-up edge or surface degradation.

8. $V_{c,min}$

After completing the qualification and ensuring that the tool is able to perform the cutting experiment, the second step is to calculate the operational domains of the tool-material combination. So, the objective of the next stage of TMP is to determine the minimum cutting speed for which the operation remains stable and efficient. A sharp increase in the slope of K_c or R_a curve, or the generation of unacceptable chips marks the minimum cutting speed.

The procedure involves utilizing the V_c from qualification tests as standard point and decreasing V_c with continuous decrements of 10% till the anomaly in any of the parameters is observed.

8.1. K_c Evaluation

As evident from equation (4), the value of K_c is independent of cutting speed. This means under stable conditions, we should not observe any variation in values of K_c by varying cutting speed. So, any significant disparity in K_c values will prove the instability of the process (usually a rise in K_c is observed with cutting speeds below $V_{c,min}$). This could be explained by formation of BUE at such speeds, caused by insignificant heat generation to effectively soften the material, while causing welding of material on surface.

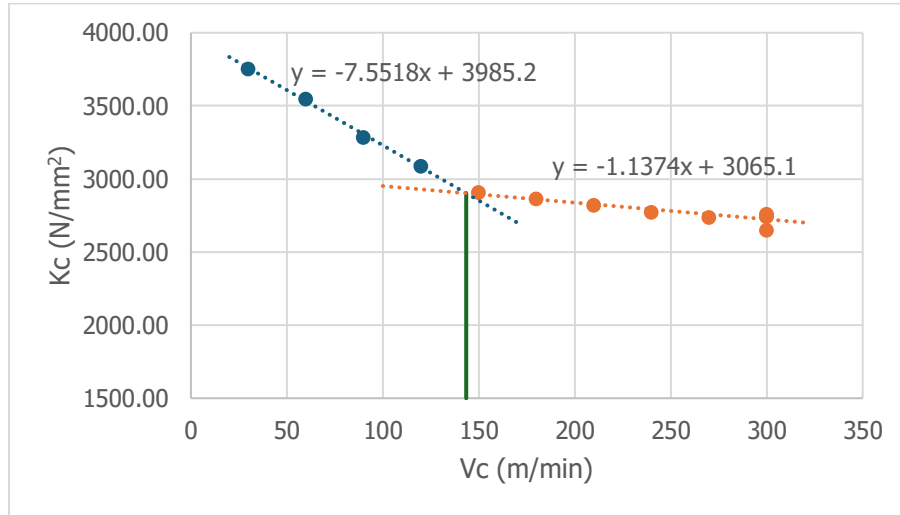


Figure 14: K_c variation with V_c decrements

It can be observed that below 150 m/min cutting speed, a significant rise in slope was observed which continued with the decrease in cutting speeds. So, by performing curve fitting to better point out the value of $V_{c,min}$,

Equations representing the best-fit lines:

$$y + 7.5518x = 3985.2$$

$$y + 71.1374x = 3065.1$$

So, through simultaneously solving both equations,

$$x = V_{c,min} = 143.4 \text{ m/min}$$

The force measurement plots on the sensor software were stable, indicating acceptable process stability.

8.2. R_a Evaluation

The experimental data provided by the previous groups was not sufficient to properly plot the evolution of R_a against the decrease in V_c to determine $V_{c,min}$. As from our K_c evaluation, it was expected to observe our $V_{c,min}$ in the range of 80-150 m/min. However, our current plot stops at 150m/min that explains our inability to comment on $V_{c,min}$. The evolution of R_a in the available test results is stable with the decrease in cutting speed, and values remained under the theoretical expectation.

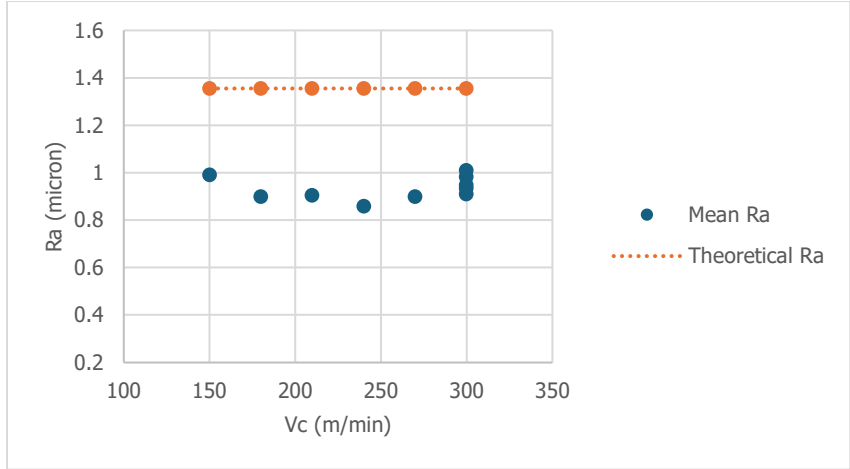


Figure 15: R_a evolution with V_c decrements

8.3. Chips Evaluation

At this stage, we were not provided with all the pictures of chips during the $V_{c,min}$ evaluation step, so it was not possible to comment on the relationship between the evolution of chips with variation in V_c . The chips provided by the previous group are tabulated below.

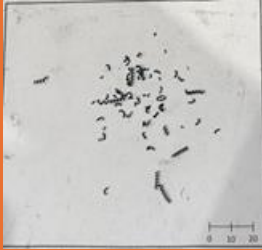
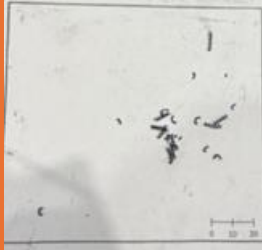
	
$V_c = 240 \text{ m/min}$	$V_c = 180 \text{ m/min}$
Category: 4.2	Category: 4.2

Table 9: Chips obtained for $V_{c,min}$ evaluation

According to Figure 11, both chips were in an acceptable state and if we focus on their color, both had a dark blue appearance. This indicates a high heat transfer rate between the insert and workpiece, depicting an excellent surface finish.

In theory, we were required to get three values for $V_{c,min}$ from three different evaluating parameters and select the maximum value from them as our experimentally determined $V_{c,min}$.

$$V_{c,min} = \text{Max}[V_{c \min(K_c)}, V_{c \min(R_a)}, V_{c \min(chips)}]$$

But due to lack of data, we could only choose

$$V_{c \min(K_c)} = 143.4 \text{ m/min}$$

9. Work Domain Determination

9.1. f_{min} and f_{max}

As shown in table (11), the chips remained in an acceptable condition over the feed rate range of $f = 0.22$ to 0.10 mm/rev. Within this range, both chip types exhibited a dark blue coloration, indicating elevated cutting temperatures and a strong heat transfer between the cutting insert and the workpiece. Such thermal conditions are generally associated with stable plastic deformation and reduced material adhesion at the tool–chip interface, thereby minimizing built-up edge (BUE) formation and contributing to improved surface finish.

Based on chip color characteristics, it is expected that unacceptable chips would display a dark brown or yellow appearance, which corresponds to lower oxidation levels and thinner oxide layers evidence of comparatively lower chip formation temperatures.

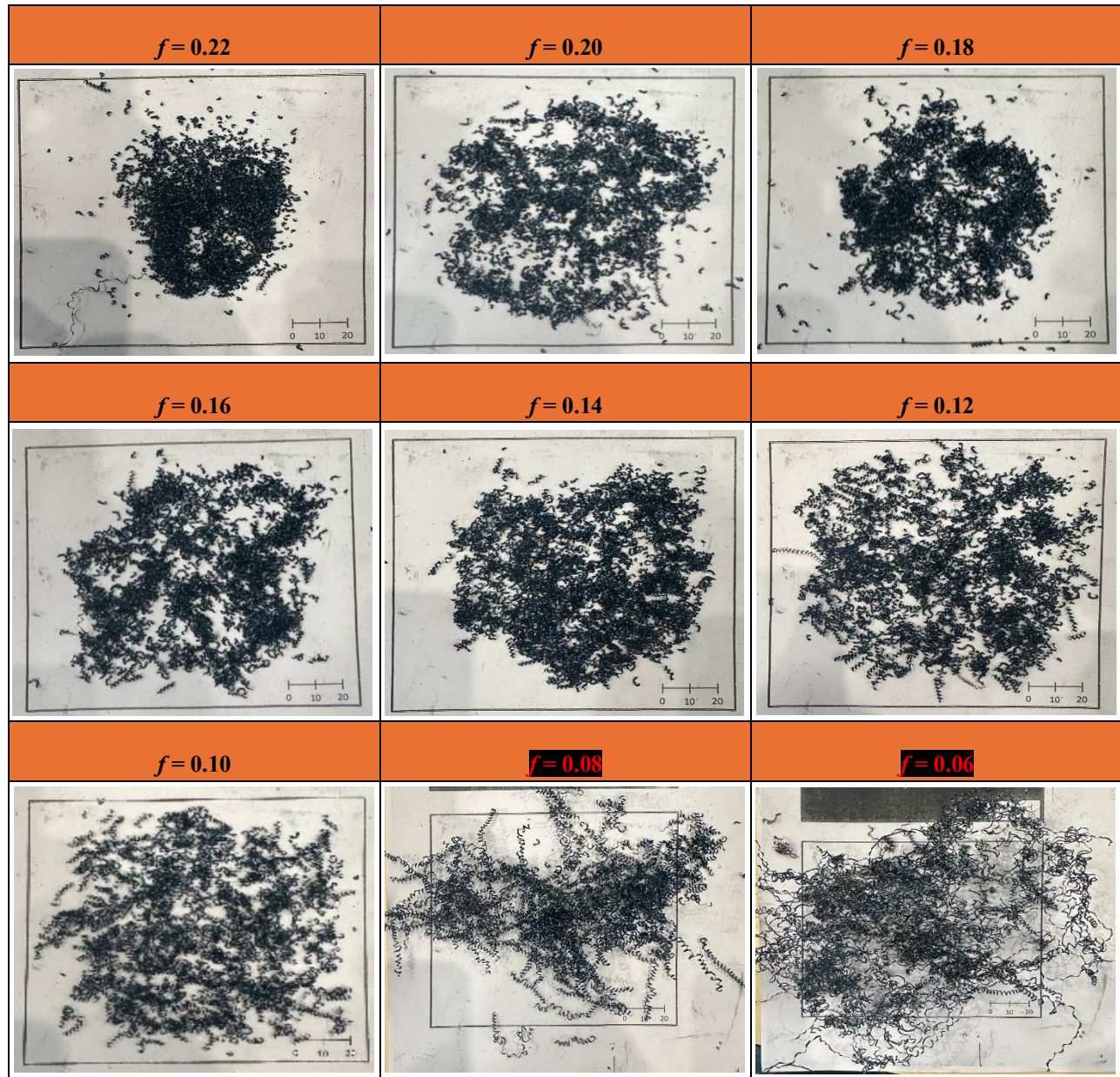
However, when the feed rate was further reduced to $f = 0.08$ – 0.06 mm/rev, the chips became long, curly, and patchy, indicating unstable chip breaking. These chip morphologies are considered unacceptable, as they can interfere with machining safety, chip evacuation, and surface integrity.

The maximum permissible feed rate is commonly defined as half of the tool nose radius ($r_e/2$). In this study, the insert nose radius was $r_e = 0.4$ mm, which corresponds to a theoretical maximum feed rate of $f = 0.20$ mm/rev.

Although feed rates approaching this limit can still be used, operating at or near $f = 0.5r_e$ places the cutting process at the boundary of the stable machining domain. At this corner of the working domain, chip formation becomes highly sensitive to small variations in cutting conditions. As a result, chips may transition from stable to unstable morphology, and surface quality can deteriorate due to increased ploughing effects, fluctuating cutting forces, and reduced chip-breaking efficiency.

Therefore, while machining at the maximum feed rate is theoretically feasible, it is generally avoided in practice, as it increases the risk of unstable chip formation and degraded surface integrity.

So, $f_{(chips)} = [0.10, 0.22]$ mm/rev



Test	Feed (mm/rev)	F_c (N)	K_c (N/mm ²)	Experimental $R_{a,ave}$ (μm)	Theoretical R_a (μm)	% Error R_a	Chip	Stability
1	0.22	266.87	2426.09	2.06	3.88	46.84	6.2	Ok (corner of the working domain)
2	0.20	256.87	2568.70	1.78	3.21	44.39	6.2	OK
3	0.18	236.68	2629.78	1.54	2.60	40.82	6.2	OK

4	0.16	215.67	2695.88	1.23	2.05	40.20		OK
5	0.14	193.90	2770.00	0.98	1.57	37.65	6.2	OK
6	0.12	172.62	2877.00	0.75	1.15	34.90	4.2	OK
7	0.10	153.72	3074.40	0.58	0.80	27.38	4.2	OK
8	0.08	131.88	3297.00	0.38	0.51	25.50	6.1	NOT
9	0.06	110.30	3676.67	0.33	0.29	12.70	5.3	NOT

Table 10: Cutting force, specific pressure, and surface roughness values obtained at different feed rates ($V_c = 300$ m/min)

9.2. R_a evaluation

From our plot, it can be easily observed that the values for $R_{a(\text{exp})}$ were always lower than the calculated values from the equation (5), the dependency of R_a on feed rate was valid. The experimental data points also exhibited a somewhat parabolic trend.

On discussion with our colleagues (Pedro Cavicchioli, Sofia Cilia and Pedro Miraglia, 2025), we were told about the article from Díaz-Salamanca et al. (2024) who observed similar trends while working with 42CrMo₄. For his experiments, he used an insert nose radius of 0,4 mm, V_c range of 76–200 (m/min) and a_p from 0,05–0,25 mm. He reported that for quenched and tempered 42CrMo₄ having 0.4mm nose radius, the measured R_a was lower than the experimental curve. The main conclusion from this paper was that although feed rate and nose radius appear as dominant factors varying R_a , the tool material interaction (elastic recovery and burnishing) were probable causes of this flatter curve.

So, regarding $f_{\min}$, we could not observe a sharp increase or any trend anomaly in our curves that could depict the value of minimum feed. The values of roughness were exhibiting ($R_a \propto f^2$) without any significant degradation in surface finish. As the previous groups didn't test below 0.06 mm/rev feed, we are taking this as our minimum feed rate from the parameter R_a . Similar to the maximum feed rate, it would be 0.22 mm/rev.

So, $f_{(R_a)} = [0.06, 0.22]$ mm/rev

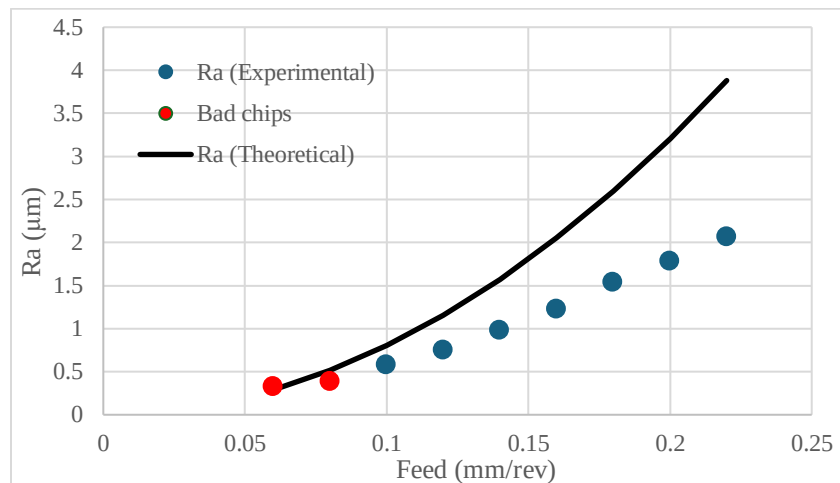


Figure 16: Plot of surface roughness against feed rate

9.3. K_c Evaluation

From the figure (, it was observed that $k_{C(\text{exp})}$ trendline demonstrated a power law decay with the increase of feed rate, like that of $k_{C(\text{theo})}$ that according to equation (1) is exhibiting ($k_c \propto 1/f$). However, a significant negative offset was observed between both curves that could be attributed to the specific metallurgical state of the workpiece (as already discussed in the qualification step). The material 42CrMo4 likely possesses a lower yield strength (annealing) in comparison to the normalized reference standard. In addition to that, thermal softening reduces the shear flow stress, thereby requiring lower energy for the operation.

From the plot, it was observed that a sharp slope increase occurred at 0.06 mm/rev. For further validation, we required more data points below 0.06 mm/rev but the experiment was stopped at this feed rate by previous groups. After discussion among our group, we decided to take 0.08 mm/rev as our minimum feed from k_c analysis.

Also, there was a slight larger dip observed at 0.22 mm/rev. It would've been better to obtain more data points beyond 0.22 mm/rev for further validation but based on this limited data and discussion with our colleagues, we settled at 0.20 mm/rev as our maximum feed.

So, $f_{(k_c)} = [0.08, 0.20]$ mm/rev

So, as we already discussed, we are going to use the *minimum of 3 values for $f_{(max)}$* and get the *maximum of 3 values for $f_{(min)}$* .

$$f_{min} = \text{Max}[f_{\min(chips)}, f_{\min(R_a)}, f_{\min(K_c)}] = [0.1, 0.06, 0.08] = 0.1 \frac{mm}{rev}$$

$$f_{max} = \text{Min}[f_{\max(chips)}, f_{\max(R_a)}, f_{\max(K_c)}] = [0.22, 0.22, 0.20] = 0.20 \frac{mm}{rev}$$

9.2. V_{cmin} and V_{cmax} determination:

From the 2nd stage of TMP, we acquired for V_{cmin} to be *143.4m/min*.

However, in order to determine V_{cmax} , it is essential to complete tool life analysis for the insert. This speed corresponds to *failure mode* of the tool, i.e., increasing the cutting speeds until the *generated thermal and mechanical stresses* cause the insert to break. It's more dependent on the *material properties, heat treatment and surface coating present*.

Based on tool life, we have to specify which *cutting time and number of parts is the manufacturer* specifically aiming for in its production process. Once specified, we can determine the average tool life of the inserts in achieving that production time and calculate V_{cmax} from that.

So, **V_c Domain = [$V_{cmin}=143.4$, $V_{cmax}= \text{unspecified}$]**

9.3. $a_{p(min)}$ and $a_{p(max)}$ determination:

For our experiment, we were not required to vary the cutting depth for the operation, it was provided to us by the manufacturer i.e., 0.3mm and was utilized for all the TMP process. Usually, $a_{p(max)}$ is provided by the manufacturer.

According to theory, $a_{p(\min)}$ is determined by decreasing a_p in specified decrements while holding feedrate constant until bad chips are observed. Similarly, the evolution of K_c and R_a is also plotted against varying a_p until a sharp slope (variation from normal trend) is observed. Thereby, the maximum value of all three $a_{p(\min)}$ is selected as the boundary limit.

10. Identification of Models

The objective of this step is to construct empirical models to expect the values of K_c and R_a based on the process inputs (cutting speed (V_c), feed rate (f), and depth of cut (a_p)) based on the equations

$$K_c = K \left(\frac{f}{f_{ref}} \right)^\alpha \left(\frac{V_c}{V_{cref}} \right)^\beta \left(\frac{a_p}{a_{pref}} \right)^\gamma \quad \text{and} \quad R_a = R \left(\frac{f}{f_{ref}} \right)^\alpha \left(\frac{V_c}{V_{cref}} \right)^\beta \left(\frac{a_p}{a_{pref}} \right)^\gamma$$

Since a_p was constant throughout the experiment, the ratio $\left(\frac{a_p}{a_{pref}} \right)$ should have no effect on K_c or R_a , and γ was considered 0. To determine the optimal empirical coefficients (K or R , α , β), a Python program was constructed to perform a non-linear regression to fit the model to the measured values. The data from qualification, $V_{c,min}$, and chip segmentation tests were the base for model fitting. However, the points with cutting speeds less than $V_{c,min}$ and points with bad chips were excluded. A key requirement of this optimization is a conservative constraint: the model is tuned to ensure that the calculated K_c or R_a is almost always greater than or equal to the measured one from test data.

- Mathematical Model

The program utilizes the following equations:

$$K_c = K \left(\frac{f}{f_{ref}} \right)^\alpha \left(\frac{V_c}{V_{cref}} \right)^\beta \quad \text{and} \quad R_a = R \left(\frac{f}{f_{ref}} \right)^\alpha \left(\frac{V_c}{V_{cref}} \right)^\beta$$

where $f_{ref} = 0.13$ and $V_{cref} = 300$

- Algorithm and Methodology:

The program is written utilizes *scipy.optimize* for numerical minimization.

- Custom Penalty Function (Conservative Fit):

Unlike standard least-squares regression (which minimizes the average error centered around the data), this program uses a custom objective function with a unilateral penalty:

1. The algorithm calculates the difference between the model output and the measured data.
2. If the model underpredicts the value, a massive numeric penalty ($10^6 \times \text{error}^2$) is added to the objective score.
3. If the model overpredicts, only the standard squared error is counted.

This forces the optimization algorithm (Nelder-Mead method) to converge on parameter values that result in a curve lying on or above the experimental data points.

10.1. Specific Pressure (K_c) Model

- **Optimized parameters:**

K	a	b
2877	- 0.2765	- 0.01467

Table 11: Optimized empirical parameters for K_c model

- **Visual representation:**

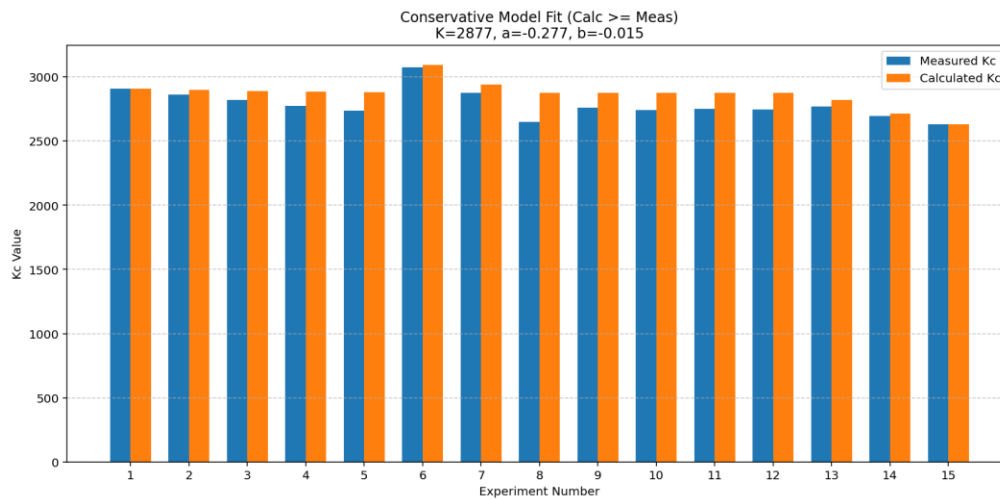


Figure 17: Plot of measured vs. calculated K_c using the model with optimized parameters

It can be observed that all calculated values are equal to or greater than the measured values, and the percentage difference ranges between 0 and 8.63%. Therefore, the model accuracy is acceptable.

10.2. Surface Roughness (R_a) Model

- **Output:**

R	a	b
1.25	0	- 0.1879

Table 12: Optimized empirical parameters for R_a model in the first run

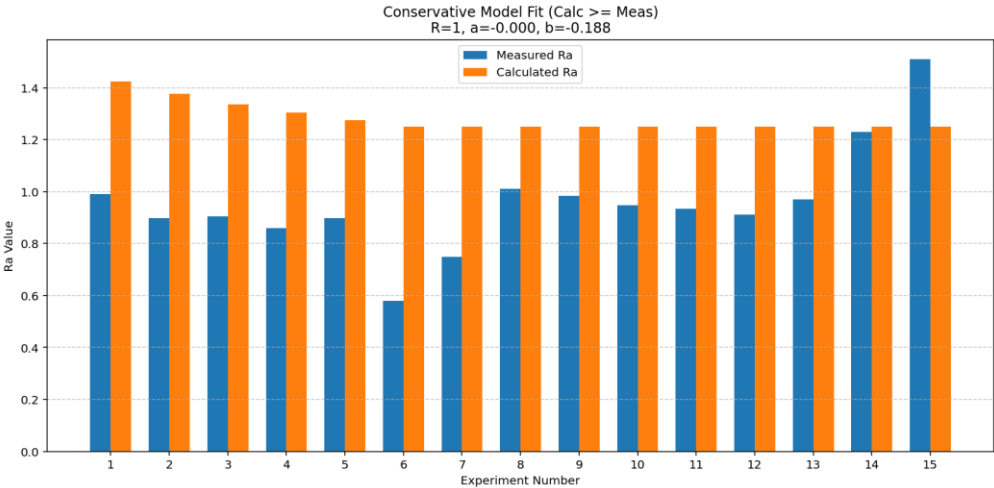


Figure 18: Plot of measured vs. calculated R_a using the model with optimized parameters in the first run

It can be noticed that the model overshoots in most of the trials with difference exceeding 100% (experiment 6) of the lab measured values, and the calculated value is less than the measured one in experiment 15. Also, the model assigns the maximum value of optimization range to R and assigns 0 to a , which means that changing the feed rate has no experimental effect on the surface roughness. These results suggest the possibility of a problem with the data fed to the model. From Table 10, the deviation of the measured R_a value from theoretical in tests 3 and 4 is a bit above 40%, which is huge; therefore, these two data points were excluded. Tests 1 and 2 were already excluded because the test resulted in bad chips.

Optimized parameters:

R	a	b
1.01	- 0.2095	0.02873

Table 13: Optimized empirical parameters for R_a model in the second run

Visual representation:

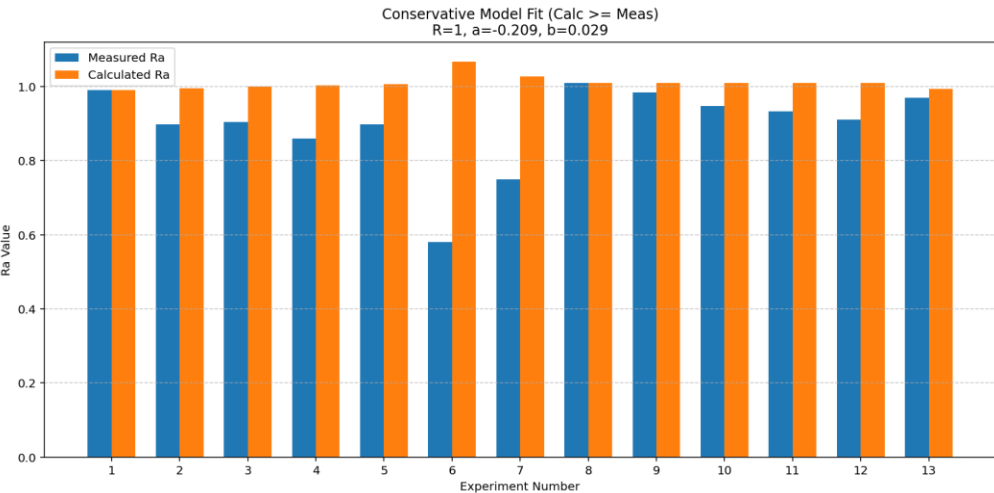


Figure 19: Plot of measured vs. calculated R_a using the model with optimized parameters in the second run

The second run yielded acceptable results with all calculated values or greater than the measured values. The tests 6 and 7 performed with feed rates 0.10 and 0.12 mm/rev achieved surface roughness values much lower than theoretical expectations, so they are expected to be much lower than the model calculations.

11. Identification of Wear Equation

Consecutive tests with a varying number of finish turning runs, and consequently time, were performed. After each test, the surface roughness of the surface, and tool wear were examined.

Test	Cumulative Time (min)	$R_{a,ave}$ (μm)	F_c (N)	K_c (N/mm ²)	Tool Wear (μm)
1	0.253	1.036	156.33	2405.08	51.38
2	0.503	1.001	183.29	2819.85	52.3
3	0.995	0.927	183.09	2816.77	64.91
4	1.714	0.862	185.81	2858.62	82.84
5	2.637	0.770	188.62	2901.85	91.26
6	3.519	0.729	190.66	2933.23	109.04
7	4.359	0.670	192.46	2960.92	120.93
8	5.158	0.636	191.73	2949.69	126.3
9	6.099	0.657	193.11	2970.92	128.6
10	7.143	0.704	195.52	3008.00	145.943
11	8.094	0.734	195.65	3010.00	151.96

Table 14: Results of tool wear tests

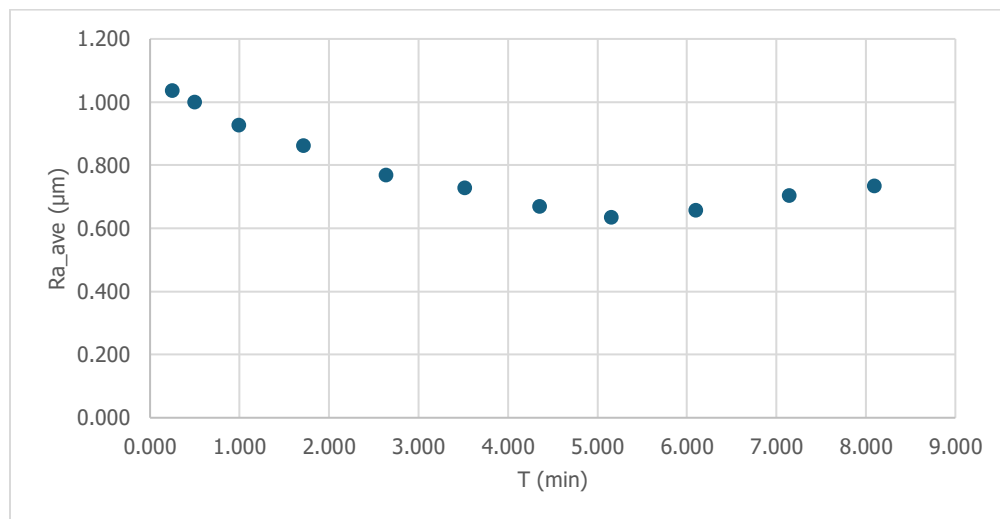


Figure 20: Plot of surface roughness over machining time

From a theoretical standpoint, surface roughness is expected to increase with machining time due to progressive tool wear and degradation of the cutting edge. However, in practical machining operations, this trend is not always monotonic. From Figure 20, similar situation observed during our experiment, the variation of average surface roughness (R_a) as a function of machining time, revealing three distinct stages in the cutting process: running-in, stabilization, and the early onset of wear.

During the initial machining period (running-in stage), comparatively higher R_a values are observed. This behavior is commonly associated with the initial interaction between the fresh cutting edge and the workpiece, where micro-irregularities at the tool edge and transient cutting conditions dominate. As the tool undergoes slight edge conditioning, these irregularities are progressively smoothed, leading to a rapid improvement in surface finish.

As machining continues, the process enters a stabilization stage, characterized by a gradual reduction followed by stabilization of surface roughness. This behavior was observed between 0 and 5.158 minutes of machining time. In this region, cutting conditions become more uniform, chip formation remains stable, and the tool-workpiece interface reaches a steady mechanical and thermal state. The relatively narrow scatter of R_a values during this phase indicates consistent surface quality and reliable machining performance.

From 5.158 min up to 8 min, a gradual increase in surface roughness is observed, which may indicate the early onset of tool wear. Due to time limitations, further extended machining and wear characterization were not conducted. However, the absence of a sharp or abrupt rise in R_a suggests that dominant wear mechanisms such as severe flank wear or built-up edge formation remain limited and do not yet significantly degrade surface integrity. This indicates that the cutting tool is still operating within its effective service life under the selected cutting parameters.

At longer machining times beyond the investigated interval, surface roughness is expected to increase more significantly as tool wear progresses. Under such conditions, wear mechanisms including pronounced flank wear and built-up edge formation would become dominant, leading to deterioration of surface quality and eventual loss of machining effectiveness.

We have also observed the value of specific pressure over machining time

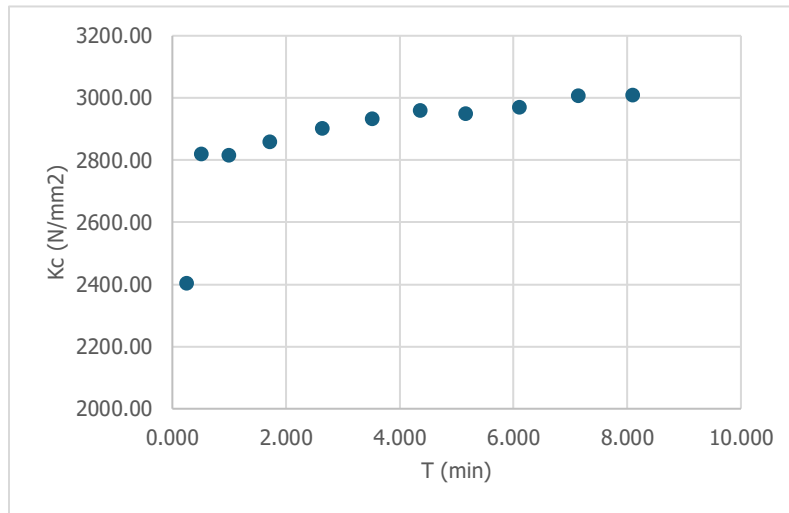


Figure 21: Plot of specific pressure over machining time

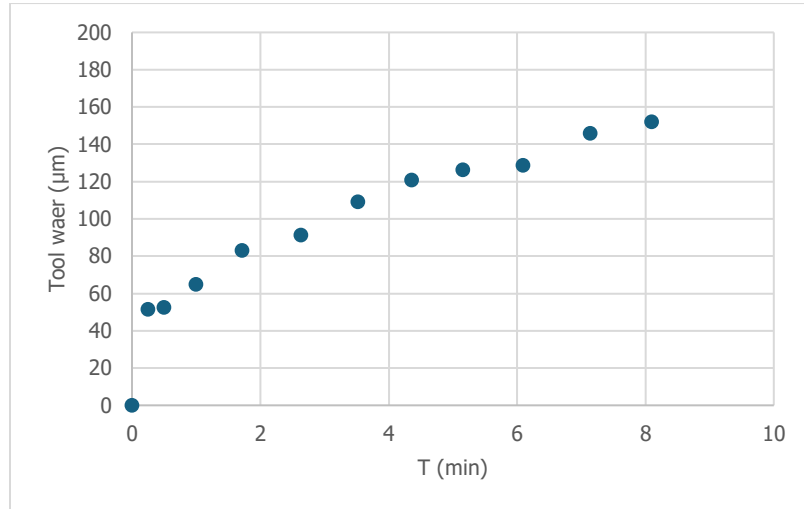


Figure 22: Evolution of tool wear with machining time

12. Conclusion

In this study, we successfully applied the Tool–Material Pair (TMP) methodology to the finish turning of 42CrMo4 steel using a CCMT 09 T3 04-PF 4415 carbide insert under dry conditions. The results show that the chosen tool, holder, and machine setup work well together, providing a stable and reliable process for finishing operations when operated within defined limits.

During the qualification stage, the process proved to meet all essential TMP criteria. Cutting forces were stable, repeatable, and stayed within $\pm 5\%$ of their average values, demonstrating consistent performance. Initially, there were some differences between the measured and handbook-based theoretical specific cutting pressure (K_c) values, highlighting the limitations of generic reference data. By using a chip-thickness-based analytical model from the Sandvik database, we achieved much closer agreement, emphasizing the importance of accounting for the actual material state and tool geometry rather than relying solely on generalized data.

Surface roughness consistently came in below theoretical predictions. This was likely due to secondary effects like elastic recovery and burnishing caused by the insert's clearance angle and nose radius. The gradual improvement in R_a during the first machining passes also reflected a normal tool run-in period, further confirming that the cutting process was stable and well-conditioned.

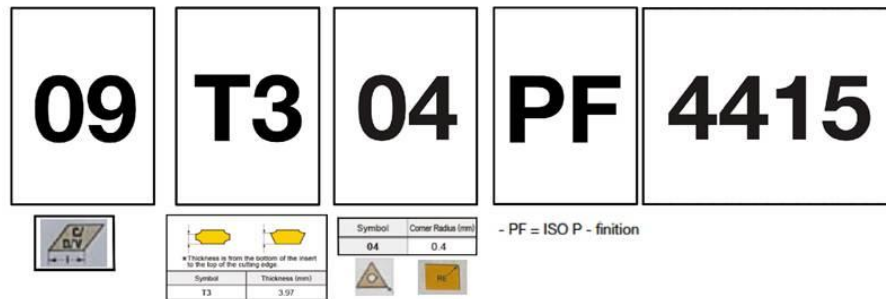
We identified the minimum cutting speed at roughly 143 m/min, where a sharp rise in specific cutting pressure marked the onset of instabilities like built-up edge formation. Feed-rate trials defined a safe and effective operating range of 0.10 to 0.20 mm/rev, balancing chip formation, surface finish, and cutting force behavior. Within this range, chips were well-formed, surfaces remained smooth, and forces behaved predictably.

Empirical models for specific cutting pressure and surface roughness were developed using a conservative approach, intentionally overestimating the measured values to ensure safe predictions for industrial use. Even with limited data at extreme conditions, these models captured the main trends and provided reliable guidance.

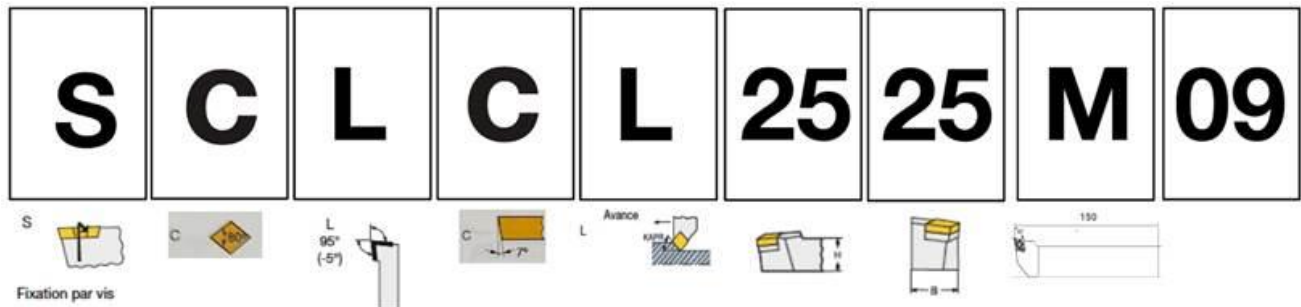
Tool wear followed a typical three-stage progression: run-in, stabilization, and early wear. Interestingly, surface roughness initially improved even as flank wear increased, showing that the tool maintained effective cutting performance throughout the monitored interval. The absence of sudden wear or surface damage confirms that the process stays well within the tool's functional limits under the chosen conditions.

In summary, the finish turning of 42CrMo4 using this tool-material pair is robust, predictable, and suitable for industrial applications when operated within the identified domain. Beyond validating this specific setup, the study demonstrates the strength of the TMP methodology as a systematic framework for understanding, modeling, and securing machining processes before full-scale production.

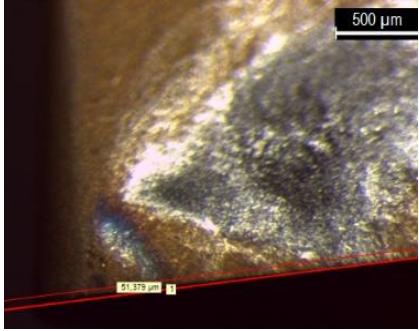
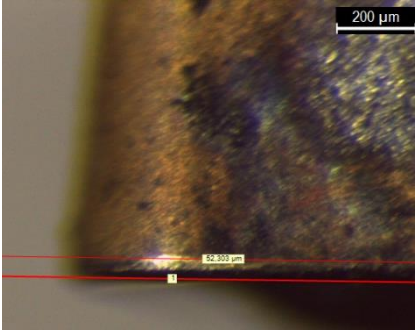
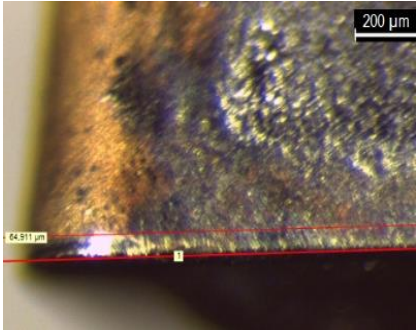
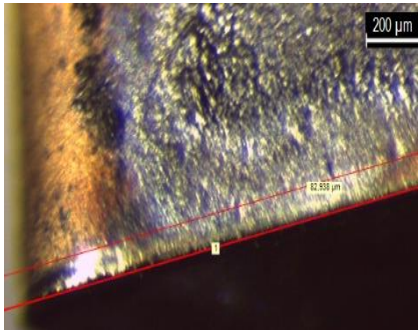
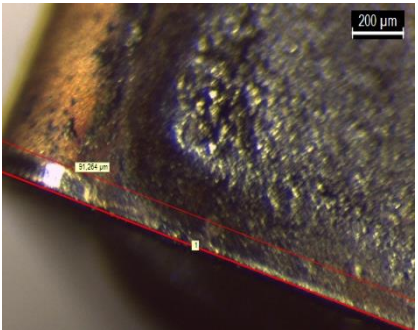
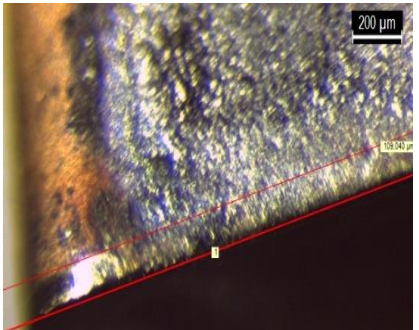
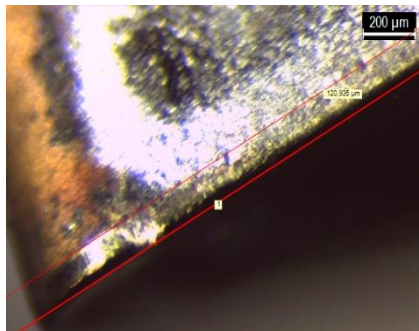
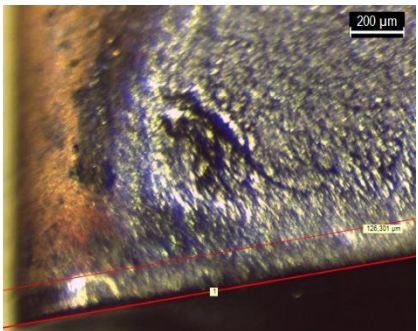
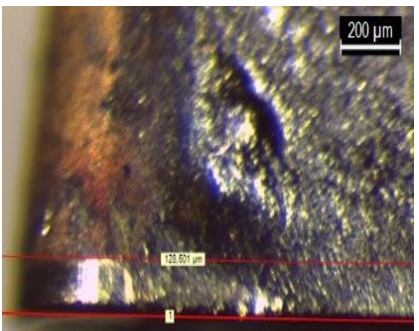
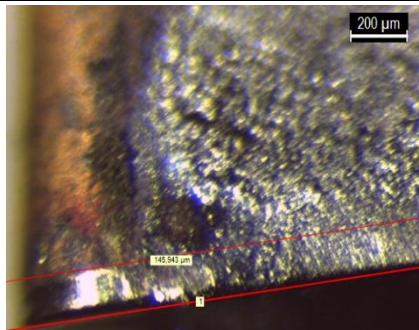
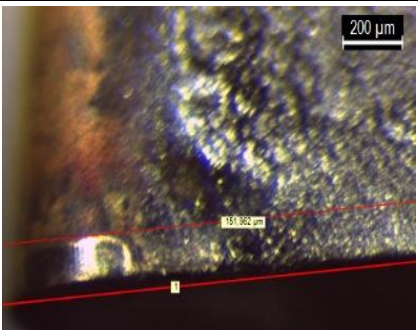
13. Appendix:



Appendix 1: Nomenclature (Second Section) of insert: 09 T3 04-PF 4415



Appendix 2: Nomenclature of Tool Holder: SCLCL 25 25 M 09

Tool wear = 51.379 μm 	Tool wear = 52.303 μm 	Tool wear = 64.911 μm 
Tool wear = 82.938 μm 	Tool wear = 91.264 μm 	Tool wear = 109.040 μm 
Tool wear = 120.935 μm 	Tool wear = 126.301 μm 	Tool wear = 128.601 μm 
Tool wear = 145.943 μm 	Tool wear = 151.962 μm 	

14. References

CAVICCHIOLI, Pedro; FIRPE, Pedro M. & CILIA, Sofia: (2025). Finish Turning

Barlier, C., & Girardin, L. (1986). *Mémotech: Productique - matériaux et usinage*. Casteilla.

Coromant, S. (2015). Technical guide: Cutting tool materials and workpiece materials [Accessed from internal PDF: Sandvik-TechnicalGuide-MaterialsISO.pdf].

Coromant, S. (2025a). Ccmt 09 t3 04-pf 4415 insert – product details. Retrieved November 23, 2025, from <https://www.sandvik.coromant.com/fr-fr/product-details?c=ccmt%2009%20t3%2004-pf%20%20%20%204415>

Coromant, S. (2025b). Ccmt 09 t3 04-pf 5015 insert – product details. Retrieved November 23, 2025, from <https://www.sandvik.coromant.com/en-gb/product-details?c=CCMT%2009%20T3%2004-PF%20%20%20%205015>

Díaz-Salamanca, D., Muñiz-Calvente, M., Ebrahimzadeh, P., Llavori, I., Zabala, A., Pando, P., Suárez-Álvarez, C., Fernández-Pariente, I., Larrañaga, M., & Papuga, J. (2024). Influence of turning parameters on residual stresses and roughness of 42crmo4+qt. *The International Journal of Advanced Manufacturing Technology*, 134, 2897–2919. <https://doi.org/10.1007/s00170-024-14257-5>